Chenrui Zhang – Statement of Purpose

Research Motivation and Vision: Computational geometry and topology provide complementary perspectives on structure, with geometry describing spatial relationships and topology capturing connectivity. Together, they enable machine learning to both represent and generate geometric and relational structure through graph-based models. I became interested in **geometric** deep learning through research and internships applying geometry- and graph-aware machine learning methods to generate and analyze structure in human motion, 3D meshes, and particle-detector data. These experiences motivated my goal of developing models that uncover relational organization and explain how predictions arise. I am drawn to questions such as: How can models that capture relational structure generalize across domains? How can models learn to generate graph or hypergraph structures that capture both local and global organization within data? How can interpretability methods reveal what modern machine learning models focus on when reasoning over irregular data? More broadly, my goal is to advance modern machine learning through geometry-aware **generative** frameworks that yield **interpretable** and reliable structure.

Research and Technical Preparation: Over the past three years, I have built a strong foundation in geometric and graph-based deep learning. I am the second author of two accepted papers (IEEE BigData 2024, AAAI MI4EGH 2024) and am preparing an ICML 2026 submission with Dr. Yusu Wang's Lab at UC San Diego. My work spans transformer architectures for motion stability analysis, benchmarking pipelines for pose estimation, and extensions of the average gradient outer product (AGOP) for interpreting GNNs. I am also leading a first-author project on generating graph structures from high-dimensional set data. These experiences strengthened my ability to design modular experiments and refine models to achieve reliable results.

In addition to research, I have practical industry experience as a Machine Learning Engineer Intern at Character Face Generation, where I applied geometric deep learning to enhance 3D mesh reconstruction quality, improving surface accuracy and visual realism in production models.

I secured the Cal-Bridge Fellowship, which provides $40,000 in flexible funding to support research or teaching during my PhD at any UC campus.

Structure-Aware Modeling in Human Motion Data: My interest in structure-aware modeling first took shape during my work on the Cal Poly Pomona Mobility Scooter Project under Dr. Tingting Chen. I analyzed stability using skeleton node features extracted from pose estimation models and aligned them with motion sensor data for kinematic analysis. I implemented deep learning models to detect instability patterns and learned that predictive performance depends critically on representing skeleton features in ways that preserve their structural relationships. This work was published at IEEE BigData 2024 and AAAI MI4EGH 2024. I realized that the success of downstream models depends on whether features are expressed in a form that preserves their **relational structure**, a principle that has guided my later research on graph-based modeling. This experience shaped my first research interest: to design models that capture relational structure and generalize reliably across diverse forms of structured data.

Learning Structure from Unordered Data: During my Cal-Bridge summer research with Dr. Yusu Wang at UC San Diego, I focused on transforming nonlinear feature spaces into structured outputs. I worked on generating cluster and relational graph representations from particle detector events. After reproducing the **Set2Graph** framework and confirming its scalability limits, I generated synthetic detector data using the Gaussian mixture model and built proximity graphs by linking each node to its 1-ring neighbors. On this foundation, I explored three modeling directions. At the node level, I developed a transformer for offset-based node generation and extended a Graph-UNet for cluster center generation. At the event level, I used cross-attention with linear assignment to generate structured graph outputs, and for clique-graph construction, I combined set2set and attention-based pairwise generation. Evaluations across fixed, dynamic, and out-of-distribution settings revealed trade-offs between clustering accuracy and structural richness. I am currently refining the models and awaiting new detector data for a full publication. This experience taught me that the challenge lies in learning structures that generalize across regimes. It shaped my interest to advance geometry-inspired methods for generating graph-structured representations from unordered features.

Interpretability in Graph Neural Networks: I continued working with Dr. Yusu Wang and began a new project applying the Average Gradient Outer Product (AGOP) to graph neural networks. AGOP is a gradient-based method that reveals how input features shape a model's representations. I trained GNN baselines on tasks such as shortest-path prediction, cycle counting, and motif detection, and generated AGOP explanations across experiments. By comparing AGOP-derived importance maps against the known ground truth and against methods such as GNNExplainer and PGExplainer, we found that AGOP provides a more direct and stable way to reveal which nodes and edges drive model predictions. Together with my collaborators, we are submitting this work to ICML 2026. Through this process, I learned how **interpretability** techniques can uncover hidden biases in graph models and that advancing GNNs requires a deeper understanding of what they implicitly prioritize during training. This experience shaped my third research interest: to advance interpretability methods that make geometric deep learning more transparent and reliable.

Applying Geometric Deep Learning to 3D Face Modeling: During my internship at Character Face Generation, I identified the limitations of the GANFit baseline and developed a mesh segmentation approach to capture finer regional detail. Building on the Mesh Transformer framework, I generated different graph representations to enhance surface reconstruction. I improved regional accuracy by incorporating edge-based pooling and curvature-aware features, achieving finer detail in areas such as the lips and eyelids while simplifying flatter regions. This experience deepened my expertise in geometric deep learning and taught me how structural representations can balance local precision with global consistency in 3D modeling.

Future Directions: Looking ahead, I plan to deepen the three research interests that have guided my work so far: modeling complex graph and topological structures, generating structured representations from unordered features, and developing interpretability methods that reveal what graph models prioritize. Together, these interests reflect my goal of creating structure-aware models that can generate and predict relational structure in data while remaining interpretable. Because many scientific datasets contain noise and hidden relationships, I aim to develop geometric and topological frameworks for structure generation and prediction that make model reasoning more transparent. My long-term goal is to advance modern machine learning through generalizable frameworks for graph **generation** and **interpretable** structured prediction across diverse domains. I also aim to mentor the next generation of researchers.